

Computer Network Design Project

Two-Floor Corporate Office Building

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Chapter 1

Wiring

The building is a two-floor corporate office complex, where the **ground floor** is served by two distribution frames: the **Main Distribution Frame (MDF)** and **Intermediate Distribution Frame IDF1**. The **first floor** is served by **IDF2** and **IDF3**. Rooms are numbered with the floor prefix: **1.x** for the ground floor and **2.x** for the first floor. Bathrooms and utility rooms (1.3, 1.9, 1.10, 1.20, 1.21, 1.28, 1.34, 1.35 and their first-floor counterparts) are excluded from network cabling.

Room types used throughout the document are defined in Table 1.1.

Table 1.1: Room Type Definitions

Type	Description	Area (m ²)	Dimensions (m)
A	Standard Office	55.25	6.5 × 8.5
B	Reception / Lobby	55.25	8.5 × 6.5
C	Individual Office	46.75	5.5 × 8.5
D	Large Conference Room	97.75	8.5 × 11.5
E	Department Office	74.75	6.5 × 11.5
F	Open-Plan Office Space	330.00	20 × 16.5
G	Large Meeting Room	200.00	10 × 20
H	Computer Lab	90.00	9 × 10
I	Seminar Room	58.50	6.5 × 9
J	Small Office	42.75	4.5 × 9.5
K	Collaborative Office	68.75	5.5 × 12.5
L	Multi-Purpose Room	70.00	5 × 14

1.1 Double Outlets

1.1.1 Calculation Methodology

Double RJ-45 Cat 6 outlets are installed in every networked room. The minimum density follows the standard of **one double outlet per 10 m²** of usable floor area, rounded up,

with a minimum of one double outlet per planned workstation position.

Each double outlet provides **two simple (single) RJ-45 sockets**.

$$N_{\text{double}} = \left\lceil \frac{A_{\text{room}}}{10} \right\rceil, \quad N_{\text{simple}} = 2 \times N_{\text{double}} \quad (1.1)$$

1.1.2 Explicit Calculation Example – Room 1.1 (Type A)

- **Room:** 1.1, Type A, Standard Office
- **Dimensions:** 6.5 m × 8.5 m
- **Floor area:** $A = 6.5 \times 8.5 = 55.25 \text{ m}^2$
- **Distribution frame:** MDF

Applying Equation 1.1:

$$N_{\text{double}} = \left\lceil \frac{55.25}{10} \right\rceil = \lceil 5.525 \rceil = \mathbf{6} \text{ double outlets}$$

$$N_{\text{simple}} = 2 \times 6 = \mathbf{12} \text{ simple outlets}$$

1.1.3 Ground Floor Outlet Summary

Table 1.2: Ground Floor – MDF Zone Outlets

Room	Type	L (m)	W (m)	Area (m ²)	Double	Simple
1.1	A	6.5	8.5	55.25	6	12
1.2	B	8.5	6.5	55.25	6	12
1.4	C	5.5	8.5	46.75	5	10
1.5	C	5.5	8.5	46.75	5	10
1.6	C	5.5	8.5	46.75	5	10
1.7	C	5.5	8.5	46.75	5	10
1.8	C	5.5	8.5	46.75	5	10
1.11	D	8.5	11.5	97.75	10	20
1.12	F	20.0	16.5	330.00	33	66
1.13	E	6.5	11.5	74.75	8	16
1.14	E	6.5	11.5	74.75	8	16
1.15	E	6.5	11.5	74.75	8	16
1.16	E	6.5	11.5	74.75	8	16
1.17	E	6.5	11.5	74.75	8	16
1.22	G	10.0	20.0	200.00	20	40
MDF Subtotal					140	280

Table 1.3: Ground Floor – IDF1 Zone Outlets

Room	Type	L (m)	W (m)	Area (m ²)	Double	Simple
1.18	E	6.5	11.5	74.75	8	16
1.19	E	6.5	11.5	74.75	8	16
1.23	G	10	20.0	200.00	20	40
1.24	H	9	10.0	90.00	9	18
1.25	H	9	10.0	90.00	9	18
1.26	I	6.5	9.0	58.50	6	12
1.27	I	6.5	9.0	58.50	6	12
1.29	J	4.5	9.5	42.75	5	10
1.30	J	4.5	9.5	42.75	5	10
1.31	J	4.5	9.5	42.75	5	10
1.32	J	4.5	9.5	42.75	5	10
1.33	J	4.5	9.5	42.75	5	10
1.36	K	5.5	12.5	68.75	7	14
1.37	K	5.5	12.5	68.75	7	14
1.38	K	5.5	12.5	68.75	7	14
1.39	K	5.5	12.5	68.75	7	14
1.40	K	5.5	12.5	68.75	7	14
1.41	K	5.5	12.5	68.75	7	14
IDF1 Subtotal					133	266

Ground Floor Total: 273 double outlets = 546 simple outlets

1.1.4 First Floor Outlet Summary

The first floor mirrors the ground floor in the IDF2 zone (same room types as the MDF zone). The IDF3 zone differs slightly: no H-type computer labs, instead, a combined L-type room is present.

Table 1.4: First Floor – IDF2 Zone Outlets (mirrors MDF zone)

Room	Type	L (m)	W (m)	Area (m ²)	Double	Simple
2.1	A	6.5	8.5	55.25	6	12
2.2	B	8.5	6.5	55.25	6	12
2.4	C	5.5	8.5	46.75	5	10
2.5	C	5.5	8.5	46.75	5	10
2.6	C	5.5	8.5	46.75	5	10
2.7	C	5.5	8.5	46.75	5	10
2.8	C	5.5	8.5	46.75	5	10
2.11	D	8.5	11.5	97.75	10	20
2.12	F	20.0	16.5	330.00	33	66
2.13	E	6.5	11.5	74.75	8	16

Table 1.4 (continued)

Room	Type	L (m)	W (m)	Area (m ²)	Double	Simple
2.14	E	6.5	11.5	74.75	8	16
2.15	E	6.5	11.5	74.75	8	16
2.16	E	6.5	11.5	74.75	8	16
2.17	E	6.5	11.5	74.75	8	16
2.22	G	10.0	20.0	200.00	20	40
IDF2 Subtotal					140	280

Table 1.5: First Floor – IDF3 Zone Outlets

Room	Type	L (m)	W (m)	Area (m ²)	Double	Simple
2.18	E	6.5	11.5	74.75	8	16
2.19	E	6.5	11.5	74.75	8	16
2.23	G	10	20.0	200.00	20	40
2.24/25	L	5.0	14.0	70.00	7	14
2.26	I	9.0	6.5	58.50	6	12
2.27	I	6.5	9.0	58.50	6	12
2.29	J	4.5	9.5	42.75	5	10
2.30	J	4.5	9.5	42.75	5	10
2.31	J	4.5	9.5	42.75	5	10
2.32	J	4.5	9.5	42.75	5	10
2.33	J	4.5	9.5	42.75	5	10
2.36	K	5.5	12.5	68.75	7	14
2.37	K	5.5	12.5	68.75	7	14
2.38	K	5.5	12.5	68.75	7	14
2.39	K	5.5	12.5	68.75	7	14
2.40	K	5.5	12.5	68.75	7	14
2.41	K	5.5	12.5	68.75	7	14
IDF3 Subtotal					122	244

First Floor Total: 262 double outlets = 524 simple outlets

Table 1.6: Building Outlet Grand Total

Distribution Frame	Double Outlets	Simple Outlets	Cable (m)
MDF	140	280	16,980.16
IDF1	133	266	12,258.04
IDF2	140	280	16,980.16
IDF3	122	244	10,899.02
Total	535	1,070	57,117.38

1.2 UTP Cable Quantity Calculation

1.2.1 Calculation Methodology

For each room, the total UTP cable needed equals the number of simple outlets multiplied by the cable length per outlet. The cable length per outlet accounts for:

- the **horizontal distance** from the distribution frame to the room (d_{DF});
- the **average cable span inside the room** to reach each outlet (l_{avg});
- a **technical reserve** of 6 m per cable (3 m at the outlet side + 3 m at the patch-panel side).

$$L_{room} = N_{simple} \times (d_{DF} + l_{avg} + 6) \quad (1.2)$$

1.2.2 Detailed Example – Room 1.1

- Simple outlets: $N_{simple} = 12$
- Distance from room to MDF: $d_{DF} = 9.4$ m
- Average cable run inside room: $l_{avg} = 14$ m
- Technical reserve: $r = 6$ m

$$L_{room\ 1.1} = 12 \times (9.4 + 14 + 6) = 12 \times 29.4 = \mathbf{352.8\ m}$$

A summary of all rooms and their cable contributions by DF zone is given in Table 1.7.

Table 1.7: UTP Cable Length per Room – MDF Zone (Ground Floor)

Room	Type	d_{DF} (m)	l_{avg} (m)	Simple	Cable (m)
1.1	A	9.40	14	12	352.80
1.2	B	16.11	5	12	325.32
1.4	C	21.82	11	10	388.20
1.5	C	27.82	11	10	448.20
1.6	C	33.53	11	10	505.30
1.7	C	39.53	11	10	565.30
1.8	C	45.24	11	10	622.40
1.11	D	0.50	11	20	350.00
1.12	F	5.00	22	66	2,178.00
1.13	E	84.64	17	16	1,722.24
1.14	E	91.78	17	16	1,836.48
1.15	E	98.78	17	16	1,948.48
1.16	E	105.92	17	16	2,062.72
1.17	E	112.92	17	16	2,174.72
1.22	G	17.50	14	40	1,500.00
MDF Subtotal				280	16,980.16

1.2.3 Total UTP Cable Quantities

The complete building cable totals are summarised in Table 1.8. The average calculated cable span across all links is approximately 53.4 m, confirming the design stays well within the Cat 6 maximum segment length of 100 m.

Table 1.8: UTP Cable Total Summary

Parameter	Value
Total cable	57,117.38 m
Technical reserve	10%
Total with reserve	62,829 m
Average cable span	≈ 53.4 m

1.3 UTP Cable Boxes

UTP Cat 6 cable is procured in 305 m boxes. The number of boxes required is:

$$N_{\text{boxes}} = \left\lceil \frac{L_{\text{total, with reserve}}}{305} \right\rceil = \left\lceil \frac{62,829}{305} \right\rceil = \lceil 205.8 \rceil = \mathbf{207} \text{ boxes} \quad (1.3)$$

Equivalently in 2 km reels:

$$N_{\text{reels}} = \left\lceil \frac{62,829}{2,000} \right\rceil = \lceil 31.4 \rceil \approx \mathbf{29} \text{ reels (2 km each)}$$

The two procurement options are summarised in Table 1.9.

Table 1.9: UTP Cable Procurement Options

Format	Length/unit Quantity		Total length (m)
Standard box (305 m)	305 m	207	63,135
Large reel (2 000 m)	2,000 m	29	58,000

1.4 Fiber Optic Cable Calculation

1.4.1 Backbone Topology

The four distribution frames are interconnected by OM4 multimode fiber optic (50/125 μm) in a **redundant ring/mesh topology**, providing multiple paths between any two DFs. The node labelling used in the cable-planning spreadsheet is:

- **A** = MDF **B** = IDF1 **C** = IDF2 **D** = IDF3
- suffix **0** = primary port; suffix **1** = secondary (redundant) port

1.4.2 Detailed Fiber Link Calculation

Table 1.10: Fiber Optic Backbone Links

Source	Target	Length	Description
A0	B0	35 m	MDF → IDF1, ground-floor backbone
B1	A1	85 m	IDF1 → MDF, redundant inter-floor path
C0	D0	35 m	IDF2 → IDF3, first-floor backbone
D1	C1	85 m	IDF3 → IDF2, redundant inter-floor path
A1	C1	5 m	MDF ↔ IDF2, adjacent riser cross-connect
D0	B0	5 m	IDF3 ↔ IDF1, adjacent riser cross-connect
A0	C0	5 m	MDF ↔ IDF2, primary riser cross-connect
D1	B1	5 m	IDF3 ↔ IDF1, primary riser cross-connect
Total (raw)		260 m	
+ 15% procurement reserve		299 m	rounded to 300 m

Detailed example: Link A0 → B0:

$$L_{\text{raw}} = 35 \text{ m} \quad (\text{measured in floor plan})$$

$$L_{\text{reserve}} = 35 \times 1.15 = 40.25 \text{ m} \approx 41 \text{ m}$$

Duplex OM4 cable is used, so both transmit and receive fibers are contained in a single cable run. The total cable procurement for the entire backbone is:

$$L_{\text{FO, total}} = 260 \text{ m} \times 1.15 \approx \mathbf{300 \text{ m}} \quad (\text{one 300 m reel of duplex OM4})$$

Each distribution frame is equipped with one **PP-72X100G-SMF** fiber patch panel (72 LC duplex ports, OM4 compatible), accommodating all backbone connections and future expansion.

Chapter 2

VLANs and IP Address Assignment

2.1 IP Addressing Scheme

The network uses a **private address space** under the block 10.1.0.0/16, NAT-ed to the public IP address 203.0.113.0. VLANs are segmented to isolate traffic types and departments. The **Management VLAN** is protected from all other VLANs by firewall policy.

Total network capacity planning:

- Total workstations: 928
- Total IP phones: 67
- Total wireless clients (AP capacity): up to 1,000
- Estimated peak network traffic: **32 Gbps**

2.2 VLAN Assignment Table

Table 2.1: VLAN and IP Address Plan

VLAN Name		Network	Gateway	Mask	Hosts
10	Management	10.1.10.0	10.1.10.1	/24	254
19	IP_Phone_Data	10.1.19.0	10.1.19.1	/24	254
20	IP_Phone	10.1.20.0	10.1.20.1	/24	254
32	Wireless	10.1.32.0	10.1.32.1	/22	1,022
40	Data_Staff	10.1.40.0	10.1.40.1	/24	254
50	Data_Guest	10.1.50.0	10.1.50.1	/24	254
100	HR	10.1.100.0	10.1.100.1	/26	62
110	Finance	10.1.110.0	10.1.110.1	/26	62
120	IT_Ops	10.1.120.0	10.1.120.1	/26	62
130	Legal	10.1.130.0	10.1.130.1	/26	62
140	Sales_North	10.1.140.0	10.1.140.1	/26	62
150	Sales_South	10.1.150.0	10.1.150.1	/26	62
160	Sales_East	10.1.160.0	10.1.160.1	/26	62

Continued on next page

Table 2.1 (continued)

VLAN Name		Network	Gateway	Mask	Hosts
170	Sales_West	10.1.170.0	10.1.170.1	/26	62
180	Mktg_Digital	10.1.180.0	10.1.180.1	/26	62
190	Mktg_Brand	10.1.190.0	10.1.190.1	/26	62
200	Frontend_Dev	10.1.200.0	10.1.200.1	/26	62
210	Backend_Dev	10.1.210.0	10.1.210.1	/26	62
220	QA_Test	10.1.220.0	10.1.220.1	/26	62
230	Logistics	10.1.230.0	10.1.230.1	/26	62
240	Purchasing	10.1.240.0	10.1.240.1	/26	62
250	Exec_Suite	10.1.250.0	10.1.250.1	/26	62
260	DevOps_Primary	10.2.0.0	10.2.0.1	/26	62
270	DevOps_Second	10.2.10.0	10.2.10.1	/26	62
280	SRE_Internal	10.2.20.0	10.2.20.1	/26	62
290	Security_Ops	10.2.30.0	10.2.30.1	/26	62
300	Training_Lab	10.2.40.0	10.2.40.1	/26	62
Total usable IP addresses					3,532

Management VLAN security note: VLAN 10 (Management) is accessible exclusively from designated management hosts. Access from all other VLANs to VLAN 10 is **blocked** at the L3 switch and enforced by ACL on the Cisco Catalyst 9410R supervisors. Remote access to all network devices (SSH) is available only through VLAN 10.

Chapter 3

Equipment Selection

For each equipment category, two options are compared. **Option 1 (Cisco)** was selected as the preferred choice, with a justification given in Section 3.10.

3.1 L2/L3 Modular Switch Chassis

Table 3.1: Switch Chassis Comparison

Specification	Option 1: Cisco Catalyst 9410R	Option 2: HPE Aruba CX 6410 v2
Chassis slots	10 line-card slots	6 line-card slots
Max access ports	480× 1 GE or 48× mGig	480× 1 GE PoE
Backplane	3.2 Tbps	1.6 Tbps
OS	Cisco IOS-XE (DNAC ready)	AOS-CX
L3 routing	Full (OSPF, BGP, EIGRP, VRF)	Full (OSPF, BGP)
Redundant supervisors	Yes (dual SUP slots)	Yes
PoE budget max	8×3 200 W PSU	4×3 600 W PSU
Unit price	\$6,100.82	\$26,334.70

3.2 Supervisor / L3 Module

Table 3.2: Supervisor Module Comparison

Specification	Option 1: Cisco C9400-SUP-1XL	Option 2: HPE CX 6400 Mgmt Module
Uplink ports	2×40 GE + 2×25 GE	2×100 GE QSFP28
Forwarding rate	480 Mpps	400 Mpps
Power consumption	350 W	200 W
HA features	In-Service Software Upgrade (ISSU)	ISSU
Management	NetConf, RestConf, YANG	REST API
Unit price	\$7,123.14	\$14,125.61

3.3 48-Port Access Line Card (Data / IP Phone)

Table 3.3: 48-Port PoE Line Card Comparison

Specification	Option 1: Cisco C9400-LC-48P	Option 2: HPE CX 6400 R0X39C
Port count	48× 1 GE RJ-45	48× 1 GE RJ-45
PoE standard	802.3at PoE+ (30 W/port)	802.3at PoE+
Max PoE budget	2,880 W	740 W
Throughput per card	96 Gbps	96 Gbps
Unit price	\$4,959.83	\$5,350.00

3.4 Fiber Optic Line Card (min 40 Gbps ports)

Table 3.4: FO Line Card Comparison (min 40 Gbps)

Specification	Option 1: Cisco C9400-LC-12QC	Option 2: HPE CX 6400 R0X45C
Port count	12× 40 GE QSFP+	8× 100 GE QSFP28
Min port speed	40 Gbps (meets requirement)	100 Gbps
Total bandwidth	480 Gbps	800 Gbps
FO module (Cisco)	QSFP-40G-SR4, \$343.64/port	HP JL310A, \$1,295/port
OM4 reach	150 m	100 m
Line card price	\$17,336.86	\$47,650.00

3.5 Multi-Gigabit AP Line Card

Table 3.5: AP Line Card Comparison

Specification	Option 1: Cisco C9400-LC-48UX	Option 2: HPE CX 6400 S1T83A
Port count	48× mGig (1/2.5/5/10 GE)	24× 2.5 GE SFP+
PoE standard	802.3bt PoE++ (90 W/port)	802.3at PoE+
Max PoE budget	2,880 W	720 W
Ideal use	WiFi 7 APs, high-density	APs up to WiFi 6E
Unit price	\$5,015.78	\$15,545.00

3.6 Firewall

Table 3.6: Firewall Comparison

Specification	Option 1: Cisco Secure Firewall 4245	Option 2: Palo Alto PA-3440
FW throughput	70 Gbps	20 Gbps
IPS throughput	50 Gbps	12 Gbps
TLS/SSL inspection	Yes (35 Gbps)	Yes (9 Gbps)
Interfaces	2×100 GE + 16×10 GE	4×10 GE + 12×1 GE
HA support	Active/Active, Active/Standby	Active/Passive
Threat intelligence	Cisco Talos	PAN Threat Prevention
Unit price	\$152,257.81	\$68,179.32

3.7 WLAN Controller

Table 3.7: WLAN Controller Comparison

Specification	Option 1: Cisco CW9800L	Option 2: HPE Aruba Central
Form factor	Physical appliance (1U)	Cloud SaaS
AP capacity	2,000 APs	Unlimited (subscription)
Throughput	160 Gbps	Cloud-based
WiFi 7 (802.11be)	Yes	Yes
On-premises operation	Yes (no Internet required)	Requires Internet
Price model	One-time \$3,686.60	Annual subscription

3.8 Access Points (min IEEE 802.11be / WiFi 7)

Table 3.8: Access Point Comparison

Specification	Option 1: Cisco CW9178I	Option 2: HPE Aruba AP-765
WiFi standard	802.11be (WiFi 7)	802.11be (WiFi 7)
Frequency bands	2.4 / 5 / 6 GHz (tri-band)	2.4 / 5 / 6 GHz (tri-band)
Max throughput	6.5 Gbps	5.76 Gbps
MIMO streams	4×4 (6 GHz), 4×4 (5 GHz)	4×4 (6 GHz)
PoE requirement	802.3bt PoE++ (60 W)	802.3bt PoE++ (30 W)
Uplink port	2.5 GE mGig	2.5 GE mGig
Unit price	\$5,455.89	\$1,475.37

3.9 IP Phones

Table 3.9: IP Phone Comparison

Specification	Option 1: Cisco IP Phone 8861	Option 2: HP 82M93AA (HP 4110)
Display	5 in colour VGA (800×480)	2.3 in colour
Wi-Fi	802.11ac dual-band	802.11ac
Bluetooth	4.2 (headset support)	No
USB	2× USB (PC/charging)	No
Lines	5 SIP lines	2 SIP lines
PoE standard	802.3at PoE+	802.3af PoE
Codec support	G.711, G.722, G.729, Opus	G.711, G.729
Unit price	\$366.20	\$194.99

3.10 Choice Justification

Technical Arguments for Cisco

1. **Unified ecosystem:** Cisco Catalyst 9000 switches, CW9178I APs, CW9800L controller, and Secure Firewall 4245 are all managed from a single *Catalyst Center* dashboard, reducing operational complexity.
2. **High-availability design:** The 9410R supports dual redundant supervisors, per-slot power-supply redundancy, and In-Service Software Upgrade (ISSU), enabling zero-downtime maintenance.
3. **Scalability:** The modular 9410R chassis allows future line-card additions without replacing the chassis investment.

Monetary Arguments for Cisco

The total cost for the Cisco option (excluding Firewall, UPS, and cooling) is **\$710,978.24**, compared to **\$791,727.64** for HPE Aruba, a saving of approximately **\$80,749.40 (10.2%)**.

Chapter 4

Environment

4.1 Rack Size Calculation and Choice

4.1.1 Rack Layout per Distribution Frame

Each DF is housed in a single **44U Open-Frame Server Rack**. The rack layout is built from the bottom (UPS) upward, with ventilation clearance rows between heat-generating equipment groups.

Table 4.1: MDF Rack Layout (44U total)

Space (U)	Position	Equipment	Model	Power (W)
1	Top	OF Patch Panel	PP-72X100G-SMF	–
13		L2/L3 Switch Chassis	Cisco Catalyst 9410R	–
		Supervisor ×2	C9400-SUP-1XL	350×2
		OF Line Card	C9400-LC-12QC	280
		WS LC ×5	C9400-LC-48P	65×5
		IP Phone LC ×1	C9400-LC-48P	65
		AP LC ×1	C9400-LC-48UX	1,630
		Fan Tray	C9410-FAN	600
		PSU ×3	PWR-C4-3200WAC	3,200×3
14		UTP Patch Panel ×7	CPPKL6ATG48WBL	–
2	Bottom	Ventilation clearance	–	–
1		Firewall	Cisco FW 4245	1,380
2		Ventilation clearance	–	–
12		UPS ×2	APC SRT10KXLT	–
44		Total		

Table 4.2: IDF1 / IDF2 / IDF3 Rack Layout (39U used of 44U)

Space (U)	Equipment	Model	Power (W)
1	OF Patch Panel	PP-72X100G-SMF	–
13	L2/L3 Switch Chassis (same config as MDF, no Firewall)	Cisco Catalyst 9410R	–
14	UTP Patch Panel ×7	CPPKL6ATG48WBL	–
12	UPS ×2	APC SRT10KXLT	–
40	Total used	5U expansion headroom	

4.1.2 Rack Options

Table 4.3: Rack Comparison

Specification	Option 1: APC NetShelter SX 44U	Option 2: Rittal TS IT 44U
Height	44U (2,033 mm)	44U (2,055 mm)
Type	Open-frame, 4-post	Enclosed, 4-post
Load capacity	1,360 kg	1,500 kg
Rail depth	Adjustable (480–900 mm)	Fixed 800/1,000/1,200 mm
Airflow	Excellent (open frame)	Managed (perforated doors)
Price per unit (USD)	≈\$800	≈\$1,200
Quantity	4	4
Total	\$3,200	\$4,800

Choice: APC NetShelter SX 44U open-frame rack. The open-frame design maximises natural airflow and reduces cooling load, is \$400 cheaper per unit, and provides easier front-rear cable access during maintenance.

4.2 UPS Power Calculation and Choice

4.2.1 Power Requirement per Distribution Frame

The required UPS power is the sum of all powered equipment in the rack, with a 30% safety margin per project specification. UPS units only supply network and server equipment; cooling units are on a separate AC circuit.

Table 4.4: UPS Power Calculation per DF

Parameter	MDF	IDF1	IDF2	IDF3
Chassis power (W)	4,030	4,030	4,030	4,030
Firewall (W)	1,380	–	–	–
Total heat (BTU/h)	17,013	13,553	13,553	13,553
Required UPS power (W)	11,115	9,600	9,600	9,600
Available UPS power (W)	20,000	20,000	20,000	20,000
Safety margin	80%	108%	108%	108%

Two **APC SRT10KXLT** units per DF give 20 kW available, which is 80–108% above the required load.

4.2.2 UPS Options

Table 4.5: UPS Comparison

Specification	Option 1: APC SRT10KXLT	Option 2: Eaton 9PX 10kVA
Rated power	10,000 VA / 10,000 W	10,000 VA / 9,000 W
Technology	Online double-conversion	Online double-conversion
Form factor	Tower / 6U rack-convertible	6U rack
Transfer time	0 ms (always online)	0 ms
Runtime (50% load)	≈15 min (expandable)	≈15 min
Management	SNMP / Web (card option)	Network-MS card
Unit price (USD)	≈\$4,500	≈\$4,200
Qty (2/DF × 4 DF)	8	8
Total (USD)	≈\$36,000	≈\$33,600

Choice: APC SRT10KXLT. Wider availability, established service network, and seamless integration with APC EcoStruxure infrastructure management platform. The 10 kW rating provides a larger power buffer, which is preferred for the MDF rack housing the firewall.

4.3 Cooling Capacity Calculation and Choice

4.3.1 Heat Dissipation per Distribution Frame

Total heat dissipated must be removed by the cooling system. A minimum 30% additional cooling capacity is required above the calculated maximum.

Table 4.6: Cooling Requirement per DF

Parameter	MDF	IDF1	IDF2	IDF3
Total heat (BTU/h)	17,013	13,553	13,553	13,553
Required (+30%) (BTU/h)	22,117	17,619	17,619	17,619
Installed cooling (BTU/h)	38,000	38,000	38,000	38,000
Actual safety margin	123%	180%	180%	180%
AC units per DF	2	2	2	2

Two **Daikin Sky Air ADEAS60A** units per DF provide 38,000 BTU/h.

4.3.2 Cooling Options

Table 4.7: Cooling Unit Comparison

Specification	Option 1: Daikin ADEAS60A	Option 2: Mitsubishi PEAD-M60JA
Type	Ducted air conditioner	Ceiling-concealed ducted
Cooling capacity	6.0 kW / 19,070 BTU/h	6.0 kW / 20,480 BTU/h
Heating capacity	7.0 kW	7.0 kW
Energy rating	A++ (SEER 7.4)	A+ (SEER 6.8)
Sound level (indoor)	28 dB(A)	30 dB(A)
Refrigerant	R-32	R-32
Electric input	1,600 W (cooling)	1,700 W (cooling)
Unit price (USD)	≈\$3,500	≈\$3,200
Total (8 units)	≈\$28,000	≈\$25,600

Choice: Daikin ADEAS60A. Higher energy efficiency reduces long-term operating costs, quieter operation, and Daikin’s VRV/VRF technology allows centralised monitoring of all 8 units.

Chapter 5

Active Equipment List

The network architecture utilizes a Cisco-centric ecosystem to ensure high-speed throughput and centralized management.

Table 5.1: Active Equipment Bill of Materials

Category	Model	Short Description	Qty	Unit Price	Total
Modular Chassis	C9410R	10-slot	4	€6,100.82	€24,403.28
Supervisor Engine	C9400-SUP-1XL	480 Mpps, L3 control plane	8	€7,123.14	€56,985.12
1G PoE Card	C9400-LC-48P	48 x 1G RJ-45 & PoE+	23	€4,959.83	€114,076.09
40G Fiber Card	C9400-LC-12QC	QSFP+ & 12 x 40GE, 4 x 100GE	4	€17,336.86	€69,347.44
mGig AP Card	C9400-LC-48UX	48 x mGig (up to 10G) & PoE++	4	€5,015.78	€20,063.12
Power Supply	PWR-C4-3200WAC	3200W AC	12	€862.39	€10,348.68
Chassis Fan Tray	C9410-FAN	Hot-swap system fans	4	€695.14	€2,780.56
Firewall	Secure Firewall 4245	70 Gbps FW & 16 x 10G, 2 x 100G	1	€152,257.81	€152,257.81
FW Fan Tray	FPR4200-FAN	FW fan modules	3	€695.14	€2,085.42
WLAN Controller	Cisco CW9800L	Up to 2,000 APs	1	€3,686.60	€3,686.60
Access Points	Wireless 9178	WiFi 7 & tri-band, tri-radio	75	€5,455.89	€409,191.75
IP Phones	IP Phone 8861	5-line & 5" VGA display, WiFi	67	€366.20	€24,535.40
OF Connectors	QSFP+	40GBASE-SR4 & SMF, LC duplex	50	€343.64	€17,182.00

Table 5.1 (*continued*)

Category	Model	Short Description	Qty	Unit Price	Total
UPS System	APC SRT10KXLT	10kVA/10kW	8	€13,285.79	€106,286.32
UPS Battery	SRT192BP2	8/10kVA	8	€1,833.79	€14,670.32
Cooling Unit	Daikin ADEAS60A	20,000 BTU/h & 5,7 kW	8	€1,996.01	15,968.08€
Active Equipment Total				€1,043,867.99	

Chapter 6

Material List

This chapter describes the passive infrastructure.

Table 6.1: Material List – Quantities and Prices

Type	Name	Characteristics	Unit Price	Qty	Value
Outlet	Double Outlet	Cat6 RJ-45, 2-port, wall-mount	€20.00	535	€10,700.00
UTP Cable	CommScope	Cat6 U/UTP, LSZH, 305 m box	€450.00	207	€93,150.00
FO Cable	OS2 Duplex	Singlemode 9/125, 300 m reel	€5,000.00	1	€5,000.00
Raceway	PVC Duct	40×40 mm, surface-mount, 2 m	€12.00	500	€6,000.00
Patch Panel	UTP Panel CPPKL6ATG48 ready WBL	48-port Cat6, 10 Gbps	€1,100.00	28	€30,800.00
Patch Panel	FO Panel PP-72X100G-SMF	Cisco 72×100G, OS2 SMF	€35,000.00	4	€140,000.00
Patch Cord	UTP 1m	Cat6, Blue, Rack-side	€18.00	1,070	€19,260.00
Patch Cord	UTP 3m	Cat6, Gray, User-side	€22.00	1,070	€23,540.00
Patch Cord	FO 1m	LC/LC Duplex, OS2 Singlemode	€25.00	16	€400.00
Organizer	Horizontal	Single-sided, cover	€150.00	24	€3,600.00
Rack	APC Rack NetShelter SX	44U Open-frame	€800.00	4	€3,200.00
Others	Wall boxes	Surface-mount	€5.00	535	€2,675.00
Others	Labels	Self-laminating, printable wrap	€40.00	20	€800.00
Others	Cable ties	Nylon 300mm, UV-resistant	€10.00	50	€500.00
Total Material Cost					€339,625.00

Chapter 7

Packet Tracer Simulation

This chapter documents the Packet Tracer simulation built to validate the logical design described in the previous chapters. Because Packet Tracer does not support the modular Cisco Catalyst 9410R chassis or the Palo Alto PA-3440 firewall, equivalent Packet Tracer device models are substituted while preserving the same logical behaviour: VLAN segmentation, inter-VLAN routing, VoIP telephony, wireless connectivity, DHCP address assignment, Internet reachability, and Management VLAN isolation are all demonstrated.

7.1 Network Topology Overview

The simulated topology follows a two-zone structure of the physical design: one **MDF** cluster and one **IDF** cluster, interconnected by a shared backbone trunk.

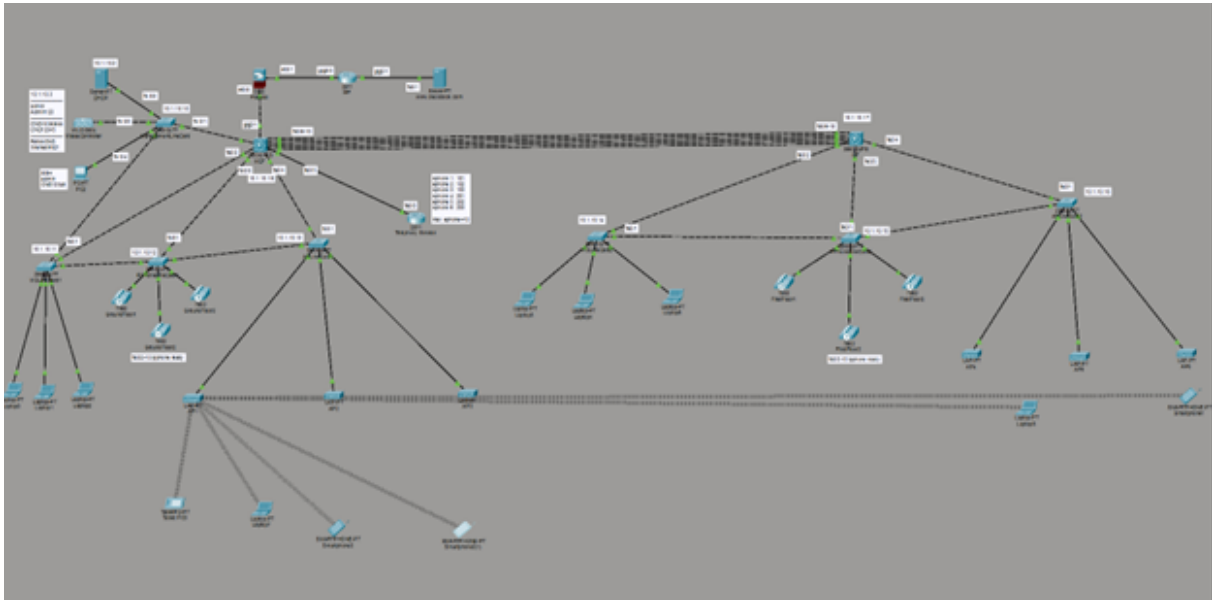


Figure 7.1: Packet Tracer – Simulated Network Topology.

Each IDF cluster is built using **data switches** (Cisco 2960-24TT) for workstations, laptops, access points and IP phones. The MDF core switch connects the IDF over a shared

backbone trunk (fa0/6–fa0/10). The firewall sits between the ISP router and the MDF core switch, enforcing the perimeter policy described in Chapter 4.

7.2 IP Addressing and DHCP

The simulation uses the **10.1.10.0/24** block for all infrastructure and end-device addresses. A dedicated **Server-PT** (10.1.10.2) acts as the centralised DHCP server, distributing addresses to all end devices (laptops, tablets, smartphones, IP phones). The default gateway for all pools points to the MDF core switch (10.1.10.18).

All end devices confirmed a successful DHCP lease before any connectivity test was performed.

7.3 VLAN Configuration and Inter-VLAN Routing

The MDF core switch (Cisco 3560) is configured with three VLANs covering the three principal traffic classes present in the building:

Table 7.1: Simulated VLANs

VLAN ID	Name	Subnet	Purpose
1	Data / Default	10.1.10.0/24	Workstations, laptops, management
10	Voice	10.1.10.0/24	IP phones (voice VLAN on phone ports)
20	Wireless	10.1.10.0/24	AP-associated wireless clients

All inter-switch links operate as **802.1Q trunks**. Phone-facing ports are configured in voice VLAN mode (`switchport voice vlan 10`), allowing Cisco 7960 phones to register with the CME router while passing data traffic on the native VLAN. Inter-VLAN routing is performed on the MDF 3560 SVIs. Connectivity between data clients, voice clients, and wireless clients was verified by `ping` tests across each pair before proceeding to feature-specific testing.

7.4 VoIP Configuration and Call Demonstration

IP telephony is implemented on a Cisco 2811 router configured as **Cisco CallManager Express (CME)**, connected to the MDF core switch. The CME is configured with six ephone-dn directory numbers supporting up to ten registered phones (`max-ephones 10`), distributed across both floors as shown in Table 7.2.

Table 7.2: IP Phone Extensions and Floor Assignment

Device	Extension	Floor	Phone Switch
7960 GroundFloor1	101	Ground	MDF
7960 GroundFloor2	102	Ground	MDF
7960 GroundFloor3	103	Ground	MDF
7960 FirstFloor1	201	First	IDF
7960 FirstFloor2	202	First	IDF
7960 FirstFloor3	203	First	IDF

A successful call between extension 101 (ground floor) and extension 201 (first floor) was demonstrated: both phones registered with the CME router, the call connected, and two-way audio was confirmed.

7.5 Wireless LAN

Six access points (AP1–AP3 on the ground floor connected to the MDF data switch; AP4–AP6 on the first floor connected to the IDF data switch) are managed centrally by the **WLC-2504** Wireless LAN Controller, which is connected to the MDF core switch. The controller broadcasts the SSID **CND-Wireless** with WPA2-PSK authentication (CND12345).

Wireless connectivity was verified on both floors:

- **Ground floor:** Laptop7, Tablet PC0, Smartphone0, and Smartphone0(1) associated with AP1–AP3 and received DHCP addresses.
- **First floor:** Laptop8 and Smartphone1 associated with AP4–AP6 and received DHCP addresses.
- All wireless clients successfully pinged wired laptops on the opposite floor, confirming end-to-end routing through the MDF core.

7.6 Internet Connectivity

A Cisco 2911 **ISP router** is connected to the WAN interface of the firewall. A web server simulating **www.ciscobook.com** is attached to the ISP router to represent an external Internet destination. The firewall performs NAT, translating internal 10.1.10.0/24 addresses to the public IP defined in Chapter 2.1 (203.0.113.6). Successful HTTP access from an internal laptop to the external web server was demonstrated, confirming that NAT and the firewall perimeter policy operate correctly.

7.7 Management and SSH Access

A dedicated **Admin PC** (PC2, credentials **admin/Admin123**) represents the only authorised management host. All switches and the CME router are configured with SSH; RSA

keys are generated on each device and login is restricted to the **admin/CND12345** local account.

Isolation was verified by confirming that a data-VLAN laptop was unable to ping the MDF core switch management address (10.1.10.18), but the Admin PC successfully opened SSH sessions to every switch and router in the topology.

Chapter 8

Network Drawings

8.1 Floor Plan – Ground Floor



Figure 8.1: Ground floor plan showing room layout, MDF and IDF1 locations, and main cable tray paths.

Key features:

- MDF room located adjacent to Room 1.11
- IDF1 located at the far wing, adjacent to Room 1.23
- Main cable tray runs along the corridor spine
- 22 networked rooms, bathrooms (1.3, 1.9, 1.10, 1.20, 1.21, 1.28, 1.34, 1.35) excluded

8.2 Floor Plan – First Floor

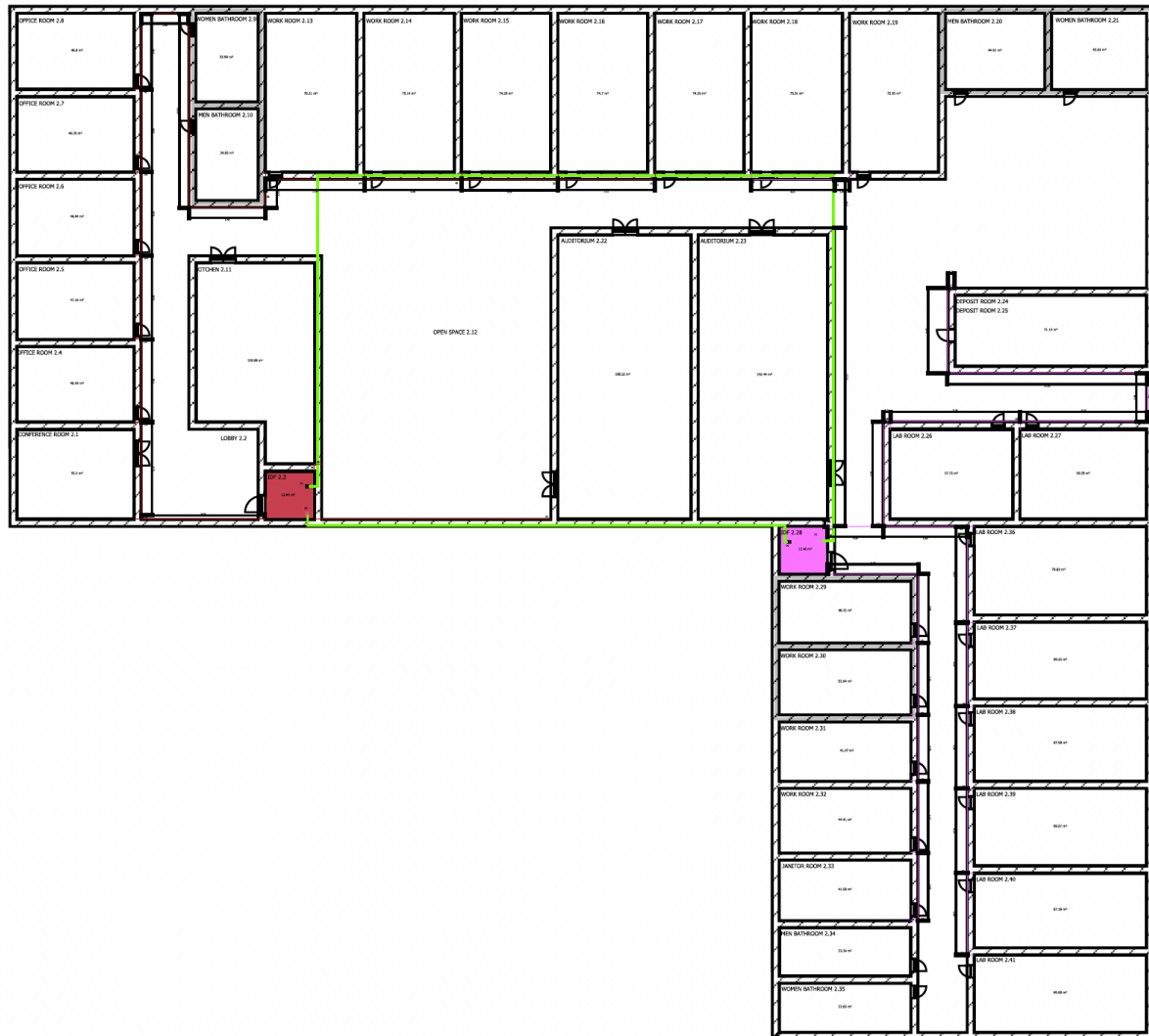


Figure 8.2: First floor plan showing room layout, IDF2 and IDF3 locations, and main cable tray paths.

Key features:

- IDF2 mirrors the MDF position on the floor above (adjacent to Room 2.11)
- IDF3 mirrors the IDF1 position on the floor above (adjacent to Room 2.23)
- Riser ducts in the building's vertical shaft connect floor DFs via FO backbone
- First floor has an additional L-type multi-purpose room (2.24/25) instead of H-type labs

8.3 Cable Path Plans

Detailed cable routing drawings with raceway paths, outlet positions, and DF locations are provided above...

Also maybe here we can include the planning of each individual room...

8.4 Network Logical Topology

```

Internet
  |
[Cisco FW 4245] -- VLAN 10 (Management)
  |
[MDF -- Cisco 9410R L3 switch / firewall uplink]
 /  |  \  \
FO  FO  FO  FO  (40GE backbone -- ring/mesh)
 /      |      \      \
IDF1  IDF2  IDF3  (via 35m + 85m + 5m runs)
 |      |      |
L2      L2      L2  (48-port PoE line cards)
 |      |      |
WS, Phones, APs per floor zone

```

8.5 Rack Diagrams

Rack layout diagrams for all four distribution frames are described in Chapter 4 (Tables 4.1 and 4.2).

Chapter 9

References

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